

Product Requirements Document: Mindbridge

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1. Overview

Mindbridge is an AI-assisted tool for psychiatrists and therapists that bridges the gap between doctors and patients by continuously tracking patient wellbeing data (EMAs, sleep, heart rate, breathing, etc.) between sessions and surfacing it as a clear, actionable clinical picture.

Instead of a doctor relying only on what a patient reports in a monthly session, Mindbridge collects passive and self-reported data over time and uses AI to summarize trends, flag anomalies, and highlight areas worth discussing — without ever diagnosing the patient.

This document covers the **prototype scope**: a doctor-only experience with simulated (fake) patient data, used to validate the core value proposition — AI-generated clinical insight reports — before building real patient-side data collection.

2. Problem Statement

Psychiatrists and therapists typically see patients infrequently (weekly to monthly) and rely heavily on the patient's memory and self-report during that short window. This creates gaps:

- Doctors miss trends that only show up over time (e.g., gradually worsening sleep).
- Patients forget or underreport symptoms between visits.
- Reviewing raw data (if collected at all) is time-consuming and not clinically formatted.

Mindbridge aims to close this gap by continuously collecting patient data and using AI to turn it into a doctor-friendly, decision-supporting summary — cutting review time and surfacing signals a doctor might otherwise miss.

3. Goals

- Validate that AI-generated summaries of patient data are genuinely useful to a doctor reviewing a patient.
- Prove out the "Generate Report" experience end-to-end using simulated data as a stand-in for real EMA/sensor data.
- Establish the data model and UX patterns that the real patient-data pipeline will later plug into.

Non-Goals (for this prototype)

- No real patient-side app or data collection.
- No diagnosis, treatment recommendations, or clinical decision-making by the AI.
- No real EMA/sensor integration (wearables, mobile sensors, etc.).
- No multi-doctor practice management, billing, or scheduling features.

4. Users

Persona	Description	Access in Prototype
Doctor (psychiatrist/therapist)	Primary user. Creates patient records, reviews AI-generated reports, uses insights to prepare for sessions.	Full access
Patient	Subject of the data; in the real product would log EMAs and connect wearables.	Out of scope for prototype (accounts may exist in the data model but have no functional login/app)

5. Scope Summary

Area	In Scope	Out of Scope
Authentication	Basic doctor login	Patient login/app
Patient creation	Doctor creates patient profile + baseline health score	Patient self-registration
Data	Simulated/fake EMA & biometric data generated from health score	Real device/EMA integration
AI Reports	Report generation, dashboard, one-page summary	Diagnosis or treatment planning

6. Feature Requirements

6.1 Authentication

- Basic username/password login.
- Only a **Doctor** role is functional in this prototype.
- Standard session handling (login, logout, session expiry).

- Patient login is explicitly out of scope but the schema should reserve a `role` field so it can be added later without rework.

Acceptance criteria:

- A doctor can log in with valid credentials and is denied with an error on invalid credentials.
 - Authenticated routes are inaccessible without a valid session.
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6.2 Doctor Dashboard

- Landing page after login.
 - Lists all patients created by/assigned to the doctor (name, username, age, health score badge, last report date).
 - Entry point to: create new patient, view patient profile, generate report.
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6.3 Patient Creation

Doctor can create a new patient record with the following fields:

Field	Type	Required	Notes
Full name	Text	Yes	
Email	Email	Yes	Used as contact; not used for login in prototype
Age	Number	Yes	
Username	Text	Yes	Unique, used to identify the patient in-app
General description	Long text	No	Free-text clinical context (e.g., presenting concerns, history)
Health score	Number, 1-10	Yes	Baseline severity/wellness indicator used to seed simulated data (1 = severe distress, 10 = thriving)
Has partner/observer	Toggle	No	If turned on, a second person (partner, family member, etc.) is attached to this patient and also logs journals/mood from their own point of view
Partner name	Text	Only if toggle is on	

When "Has partner/observer" is on, the system also seeds a second, separate stream of journals and mood logs written from the partner's perspective about the same time period.

This is useful because what the partner reports and what the patient reports can differ, and that difference (discrepancy) is itself useful data for the doctor — e.g. patient says they're fine, partner's log says they've been withdrawn all week.

Acceptance criteria:

- Form validates required fields and unique username before submission.
 - On save, a patient record is created and simulated data generation is triggered (see 6.4).
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6.4 Simulated Data Generation Engine

Since real EMA/sensor data isn't available in the prototype, the system generates a plausible synthetic dataset for each patient, driven by their **health score**.

Purpose: Give the AI report generator realistic-looking longitudinal data to summarize, so the reporting experience can be evaluated without waiting on real-world data collection.

Behavior:

- On patient creation (or on demand via a "Regenerate data" action), the system generates a time series (e.g., 30 days) covering:
 - Sleep duration & quality
 - Resting heart rate / heart rate variability
 - Breathing rate
 - EMA-style mood/affect check-ins (e.g., mood rating, anxiety rating, energy level)
 - **Mood logs** — short, structured mood entries (e.g., mood rating + 1-2 tags) simulating quick in-app check-ins
 - **Journals** — short free-text paragraph entries simulating what a patient would actually type into the app
- Health score maps to data severity. Lower health score → more disrupted metrics (e.g., shorter/fragmented sleep, elevated resting heart rate, lower mood scores, more missed check-ins, more negative journal/mood content). Higher health score → stable, healthy-range metrics and neutral-to-positive journal/mood content.
- Data should include natural noise/variance and occasional "spikes" or "dips" (not perfectly smooth) so the AI has something meaningful to detect and the dashboard has something to visualize.
- A documented mapping table (e.g., health score 1-3 = "high severity" ranges, 4-6 = "moderate," 7-10 = "stable") should back the generation logic so it's tunable.

Journal generation approach (prototype-specific):

- To avoid burning API calls generating journal text per patient, journal entries are **not** LLM-generated at seed time. Instead, the system pulls from a **pre-written, hardcoded pool of short journal snippets**, bucketed by severity/theme (e.g., breakup, substance use, work stress, sleep complaints, positive/neutral day, social withdrawal, etc.).
- Each bucket is tagged to a severity range so it lines up with the patient's health score (e.g., low health score pulls more heavily from breakup/substance-use/isolation-type snippets; high health score pulls from neutral/positive snippets).
- Entries are randomly selected (with some repetition/variation allowed) and distributed across the generated date range alongside the mood logs, so journals and mood/biometric dips can plausibly correlate on the same days.
- This snippet pool should be easy to expand later or swap for real LLM-generated or real patient text without changing the data model.

Partner/observer data:

- If the patient has "Has partner/observer" turned on, generate a second set of journal entries and mood logs, same date range, but written from the partner's point of view (own snippet pool, tagged for partner perspective).
- The partner's data doesn't have to match the patient's severity exactly — sometimes make it line up, sometimes make it differ on purpose (e.g. patient's mood log says fine, partner's journal says worried/withdrawn). This mismatch is what the doctor will see flagged later.

Acceptance criteria:

- Every created patient has a populated dataset — including journals and mood logs — before a report can be generated.
- If a partner/observer is attached, their journal/mood data is also generated for the same date range.
- Data generated (including which journal snippets get selected) is internally consistent with the assigned health score (spot-checkable).
- Doctor can trigger regeneration of the synthetic dataset for a given patient (useful for demoing different severities).

6.5 EMA Compliance

Compliance is just a measure of how consistently the patient is logging. If there are gaps in the expected frequency of mood logs, journals, or other check-ins, that lowers the compliance score. This applies later for real patient data too, but for now it can just be calculated off the simulated data (i.e. we can seed some days with missing entries on purpose so there's something to calculate).

- Each patient gets a compliance score (e.g. 0-100%), based on how many expected check-ins were actually logged over the tracked period vs how many were missing.
 - Gaps should show up somewhere on the patient profile/dashboard, not buried.
 - Not a priority for this build, just needs to be kept in mind in the data model so it's not a rework later (e.g. simulated data generator should be able to skip days on purpose, and there should be a place to store/display the resulting score).
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6.6 Generate Report (AI Analysis)

The core feature. From a patient's profile, the doctor clicks "**Generate Report.**"

Behavior:

- All available patient data (profile description + full simulated time series, **including mood logs and journal entries**) is sent to an LLM (OpenAI or Gemini, prompt-tuned for this prototype — model swap should be configurable, not hardcoded).
- The prompt explicitly instructs the model to:
 - **Never diagnose** or suggest a specific clinical diagnosis.
 - Summarize overall patterns and trends across the tracked window.
 - Identify and explain notable spikes/dips/anomalies in the data.
 - Surface correlations worth the doctor's attention (e.g., "sleep dropped sharply the same week mood scores fell, coinciding with a journal entry referencing a breakup").
 - Suggest **actionable, non-diagnostic** discussion points or monitoring suggestions (e.g., "consider asking about recent life stressors around July 10-14").
 - Keep tone clinical, neutral, and doctor-facing (not patient-facing language).
- Output is structured (not just prose) so it can populate a dashboard: summary text, list of spikes/anomalies with dates, list of insights, list of actionable suggestions, and data series for charts.

Acceptance criteria:

- Report generation completes and renders within a reasonable time with a visible loading state.
 - Output never contains diagnostic language (should be checked via prompt design and, ideally, a lightweight guardrail/keyword check).
 - Report is persisted and associated with the patient (viewable again later, not just generated live each time).
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6.7 Report Dashboard

A visually polished, doctor-facing dashboard displaying the generated report. Should include:

- **Summary section** — plain-language overview of the review period.
- **Graphs/visualizations** — sleep, heart rate, breathing, mood/EMA trends over time, clearly annotated.
- **Journal & mood log timeline** — a chronological view of journal snippets and mood logs, ideally overlaid or cross-referenced with the biometric graphs so the doctor can see what the patient wrote alongside what the data shows.
- **Partner comparison view** (if partner/observer exists) — patient's journal/mood log next to the partner's for the same days, with any noticeable mismatch called out (e.g. patient reports fine, partner reports withdrawn).
- **Compliance score** — shown somewhere visible on the dashboard, with the specific gaps/missed days listed.
- **Spikes/anomalies** — visually flagged on the relevant graph(s) and listed with dates and brief explanations, cross-referenced against journal/mood log entries from the same period when relevant.
- **Insights** — key observations pulled out from the data.
- **Actionable insights** — concrete, non-diagnostic suggestions for the doctor's next steps or session focus.
- Clear visual distinction between "data-driven" content and "AI-generated interpretation," and a note that this is not a diagnostic tool.

Acceptance criteria:

- Dashboard is readable at a glance (a doctor should be able to grasp patient status in under a minute).
 - All AI-derived content is visually distinguishable from raw data.
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6.8 One-Page Summary

A dedicated button/action: "**One Page Summary.**"

Behavior:

- Condenses the *already-generated* full report/dashboard content into a single-page, print/export-friendly summary — not a new independent AI analysis.
- Includes: headline summary, top 2-3 spikes/insights, top actionable suggestions.
- Should be easy to export (e.g., PDF) or view in a clean printable layout, since doctors will likely want this on hand during a session or for records.

Acceptance criteria:

- One-page summary is generated (or derived) from the existing report content, not a fresh AI call against raw data.
 - Fits on a single printed/exported page.
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7. Data Model (High-Level)

- **Doctor:** id, name, email, username, password (hashed)
 - **Patient:** id, doctor_id, name, email, age, username, description, health_score, has_partner, compliance_score, created_at
 - **Partner:** id, patient_id, name
 - **SimulatedDataPoint:** id, patient_id, date, sleep_hours, sleep_quality, resting_hr, hrv, breathing_rate, mood_score, anxiety_score, energy_score
 - **MoodLog:** id, patient_id, author (patient or partner), date, mood_rating, tags[]
 - **JournalEntry:** id, patient_id, author (patient or partner), date, snippet_id (ref to JournalSnippetPool), text
 - **JournalSnippetPool** (static/hardcoded): id, severity_bucket, theme (e.g., breakup, substance use, work stress, sleep complaint, neutral/positive), perspective (patient or partner), text
 - **Report:** id, patient_id, generated_at, summary_text, insights[], spikes[], actionable_insights[], chart_data (ref to data range used), model_used
 - **OnePageSummary:** id, report_id, content
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8. AI/Prompting Requirements

- Model-agnostic integration layer (OpenAI/Gemini interchangeable via config).
 - System prompt must hard-constrain the model to a **non-diagnostic, summarization/insight-support role**.
 - Output should be requested in a structured format (e.g., JSON) to reliably populate the dashboard.
 - Include the patient's free-text description and health-score context in the prompt so the AI's tone/insights are appropriately calibrated to the case.
 - Journal entries and mood log tags should be passed in with their dates so the model can correlate them against the same-day biometric/EMA data when identifying spikes and insights.
 - If a partner/observer exists, pass in both streams labeled by author so the model can point out days where the patient's and partner's reports don't match.
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9. Success Criteria for Prototype

- A doctor can go from "create patient" to "review AI report" in one sitting, with no manual data entry beyond the initial patient form.

- Reports feel specific to the patient's data, not generic boilerplate text.
 - No diagnostic language shows up in generated output.
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10. Assumptions

- Basic auth = plain username/password login, no OAuth/SSO.
 - Simulated data window is ~30 days.
 - One-page summary is a condensed version of the already-generated report, not a separate AI call.
 - Reports are saved as history (multiple reports over time per patient), not overwritten.
 - Journal entries come from a small, hand-written, hardcoded snippet pool, not LLM-generated, to save on API cost. Snippet pool content itself still needs to be written.
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11. Open Questions

- Should doctors be able to edit a patient's health score after creation, and should that regenerate the dataset?
- Should reports be exportable/shareable (e.g. PDF) beyond just the one-page summary?
- How many partner snippets/themes do we want at launch, and should they mirror the patient snippet themes or be their own set?